
Analyzing Factors Affecting POLYITAN-1 Telemetry Demodulation

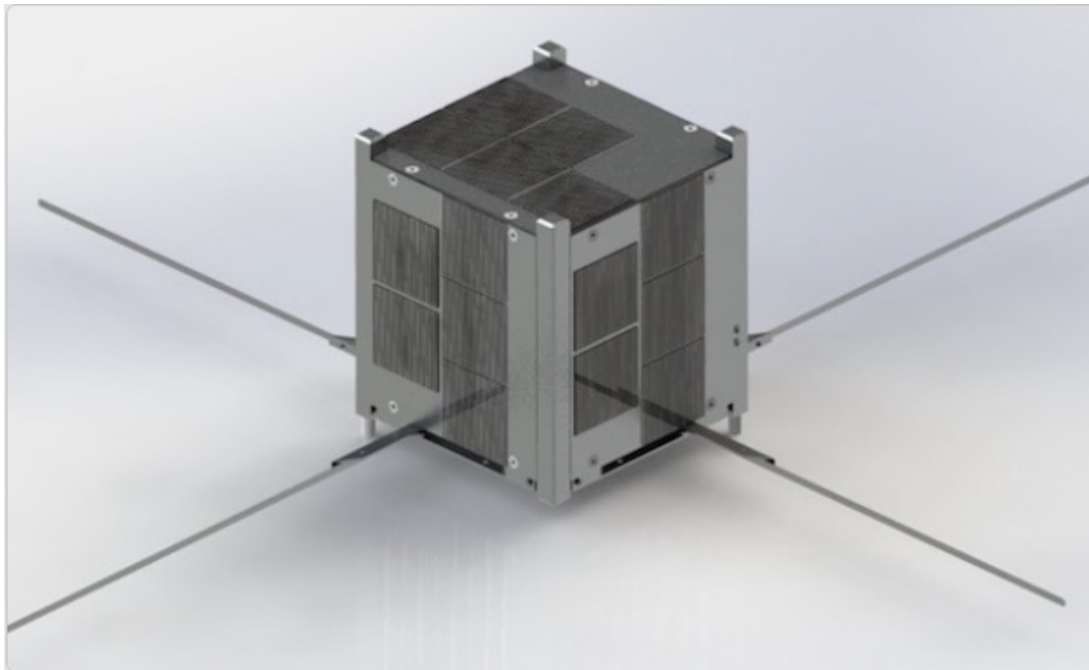


Figure 1: *The POLYITAN-1 satellite as it would be deployed on orbit[1]*

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Introduction

POLYITAN-1 is a Ukrainian cube satellite that was placed into orbit in 2014 as a technology demonstration satellite[2]. The satellite network of ground stations (SatNOGS) is a community of amateur radio operators with hundreds of satellite ground stations and has made approximately 11,000 attempts at demodulating radio signals from POLYITAN-1. However, the community has only achieved a success rate of 47.55%[5]. At this point in time, it is unclear as to why the community is having such a difficult time demodulating radio signals from this satellite.

It is likely that the elevation angle (that is, the degrees off the horizon from the perspective of the observer), whether the satellite is illuminated, since illumination means the satellite should have an abundance of power for transmission, and whether the observer is being illuminated due to increased atmospheric interference and receiver thermal noise are key factors that affect whether the radio signal is demodulated. An observation study will be conducted to assess whether transmission rate is highest when the elevation angle is high, the satellite is illuminated, and the ground station is not illuminated.

The Data

The SatNOGS community makes their data available under the creative commons license and stipulates that credit must be given for the data source and does not place any other restrictions on the data use[4]. A python script was used to pull approximately 15,000 observations from the SatNOGS website. Then, to enrich the data, the python skyfield package was used to determine the highest elevation in a given pass, whether the satellite was illuminated by the sun during the pass, and whether the ground station was illuminated by the sun during the pass[3].

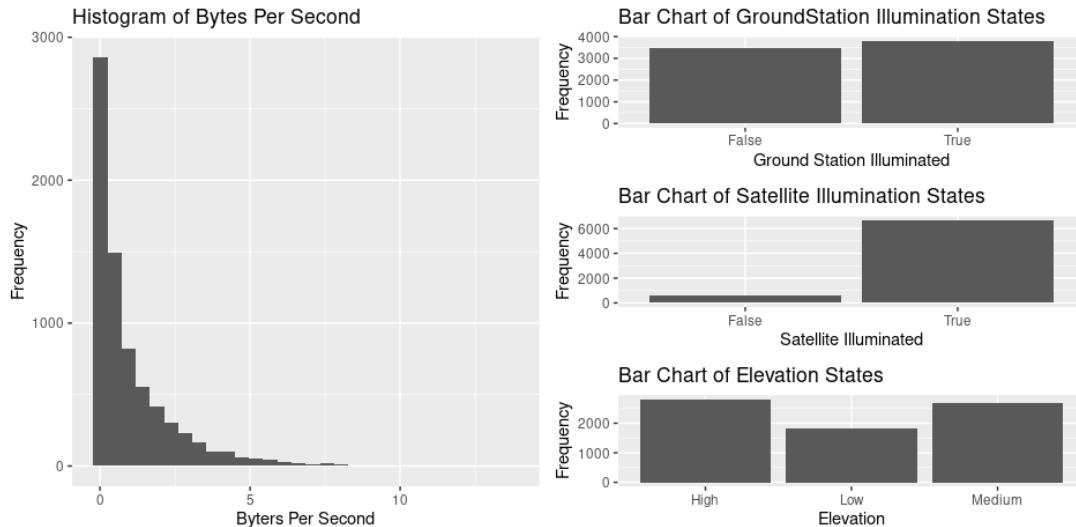


Figure 2: Plot depicting a histogram of the data collected by various ground-stations, as well as, the number of each qualitative variable class

This is specifically how the data will be referred as the following during the study:

Response : The transmission speed in bytes per second.

Factor A : Elevation Angle, levels: Low, Medium High

Factor B : Satellite Illumination State, levels: True, False

Factor C :Receiver Illumination State levels: True, False

Additionally, as seen in Figure 2, the data has strong right skew that will need to be transformed to ensure all assumptions of the model are met. Additionally, the collected data has strong class imbalance where very few of the observations have the satellite in darkness.

Method

To control for the variation between the various ground station hardware and interference environments, the ground station ID will be utilized as a blocking factor. The intention is to implement Randomized Complete Block Design (RCBD). However, if the data is too sparse, then a switch to Completely Randomized Design (CRD) will happen. Assuming there is sufficient data for RCBD, the model used for this study shall be:

$$Y_{ijklm} = \mu + \alpha_i + \beta_j + \gamma_k + \delta_l + (\alpha\beta)_{ij} + (\alpha\gamma)_{ik} + (\beta\gamma)_{jk} + (\alpha\beta\gamma)_{ijk} + \epsilon_{ijklm}$$

Where:

- μ is the mean quantity of data demodulated,
- α_i is fixed effect of the elevation angle
- β_j is the fixed effect of the satellite illumination state
- γ_k is the fixed effect of the ground station illumination state
- $\delta_l \stackrel{iid}{\sim} N(0, \sigma_\delta^2)$ is the random error associated with the the ground station block.
- $(\alpha\beta)_{ij}$, $(\alpha\gamma)_{ik}$, $(\beta\gamma)_{jk}$, and $(\alpha\beta\gamma)_{ijk}$ are the interaction effects between the elevation angle, the satellite illumination, and the ground station illumination.
- $\epsilon_{ijklm} \stackrel{iid}{\sim} N(0, \sigma_\epsilon^2)$ is the random error associated with the observation.

Results

It became clear early in the analysis that additional steps needed to be taken to prepare the data. To implement RCBD, the data was subsetting such that only observations from stations were kept only if the station appeared more than 10 times in the data. This led to the data consisting of 14829 observations from 77 ground-stations. Additionally, the transmission rate needed to undergo transformation to meet the assumptions of ANOVA, as evidenced by Figure 6 of the appendix. The box-cox method was used in SAS to find a λ value of zero (depicted in Figure 3), suggesting that the transmission rate should be transformed by taking the logarithm.

Following this transformation, the Q-Q plot is significantly healthier and the “megaphone” affect is drastically reduced, as seen in 3. The residuals versus fits plot has a band of approximately 3.5 on the top but has a line with a negative slope along the bottom. While this is a violation of the assumption of equal variance, this ANOVA study consists of 14829 observations and the conclusions should still be relevant despite the slight violation.

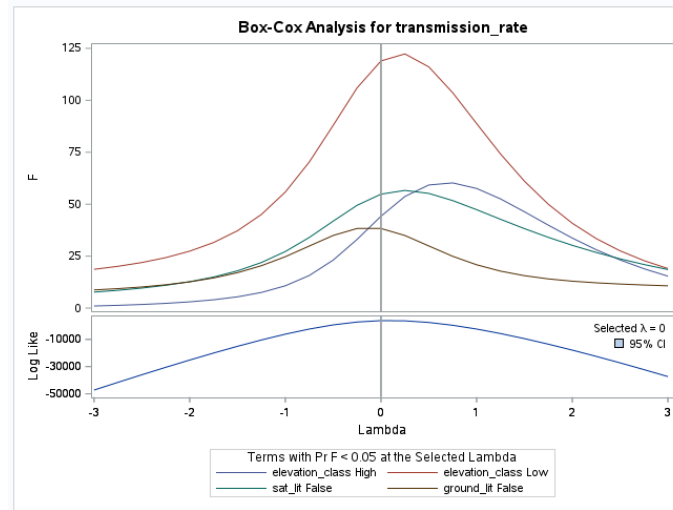


Figure 3: Using the box-cox method in SAS to discover a λ value of 0 suggesting a logarithmic transformation.

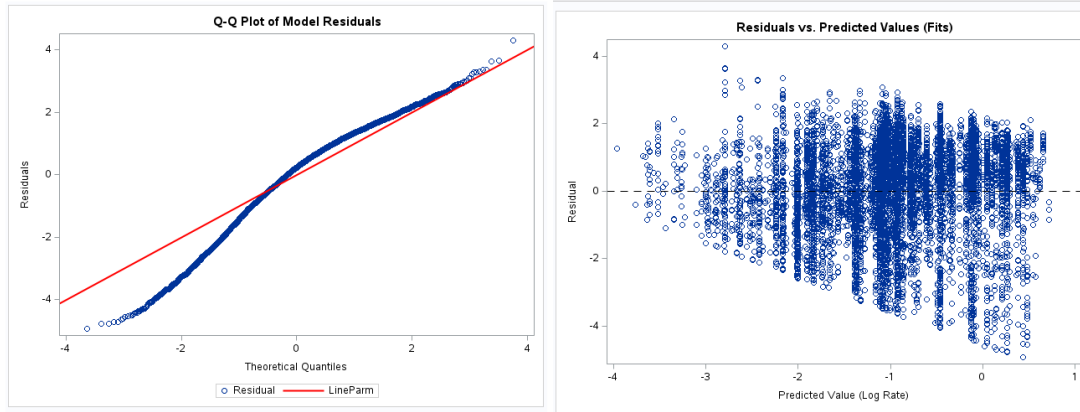


Figure 4: Q-Q Plot and Residuals vs Fits plots with the transformed transmission rate, showing significant improvements

Figure 7 of the appendix captures the full ANOVA table and revealed a problem with the collected data. Due to the satellite's low orbit, most of the observations had matching satellite illumination and ground-station illumination, resulting in very significant col-linearity. This also explains why the Sum of Squares were very similar for the both terms. Due to the col-linearity, the decision was made to reduce the model by dropping the satellite illumination variable. The reduced model ANOVA table is captured in figure 5 and the updated model with the same indices established earlier is:

Type 3 Analysis of Variance								
Source	DF	Sum of Squares	Mean Square	Expected Mean Square	Error Term	Error DF	F Value	Pr > F
elevation_class	2	741.848049	370.924025	Var(Residual) + Q(elevation_class,elevation*ground_lit)	MS(Residual)	7217	188.08	<.0001
ground_lit	1	39.303778	39.303778	Var(Residual) + Q(ground_lit,elevation*ground_lit)	MS(Residual)	7217	19.93	<.0001
elevation*ground_lit	2	9.291570	4.645785	Var(Residual) + Q(elevation*ground_lit)	MS(Residual)	7217	2.36	0.0949
Station	75	4367.205658	58.229409	Var(Residual) + 90.437 Var(Station)	MS(Residual)	7217	29.53	<.0001
Residual	7217	14233	1.972205	Var(Residual)	.	.	.	.

Figure 5: Updated ANOVA table without the co-linearity using the updated model.

$$Y_{iklm} = \mu + \alpha_i + \gamma_k + \delta_l + (\alpha\gamma)_{ik} + \epsilon_{iklm}$$

Evaluating α_i

The coefficient α_i represents the fixed effect of the elevation angle and was found to differ meaningfully from zero:

$$H_0 : \forall_i[\alpha_i = 0]; \quad H_a : \exists_i[\alpha_i \neq 0]$$
$$F^* = \frac{370.924}{1.972} = 188.1$$

The p-value of $F_{2,7217}188.1 < 0.001$, therefore reject the null hypothesis. Figure 8 of the appendix captures the diffogram and tukey means groupings, showing that high elevation observations have the highest transmission rates while low elevation observations have the lowest transmission.

Evaluating γ_k

The coefficient γ_i represents the fixed effect of the ground-station illumination and was found to differ meaningfully from zero:

$$H_0 : \forall_k[\gamma_k = 0]; \quad H_a : \exists_k[\gamma_k \neq 0]$$
$$F^* = \frac{39.30}{1.972} = 19.93$$

The p-value of $F_{1,7217}19.93 < 0.001$, therefore reject the null hypothesis. Figure 9 of the appendix captures the diffogram and tukey means groupings for ground illumination, the transmission rate is higher in darkness.

Evaluating $(\alpha\gamma)_{ik}$

The coefficient $(\alpha\gamma)_{ik}$ represents the interaction effect of the elevation angle and the ground-station illumination and was not found to differ meaningfully from zero.

$$H_0 : \forall_{i,k}[(\alpha\gamma)_{ik} = 0]; \quad H_a : \exists_{i,k}[(\alpha\gamma)_{ik} \neq 0]$$
$$F^* = \frac{4.646}{1.972} = 2.36$$

The p-value of $F_{2,7217}2.36 \approx 0.0949$, therefore fail reject the null hypothesis.

Conclusion

In conclusion, it was discovered that the elevation angle of the ground-station relative to POLYITAN-1 and whether the ground-station is illuminated have a significant effect on the demodulation of radio signals from the satellite. This conclusion is reasonable as higher elevation passes would have less obstruction from objects, such as trees or buildings. Additionally, the finding higher transmission rates is reasonable due to less interference in the atmosphere and thermal noise on the receiver from the sun. It follows from these observations that the SatNOGS community could achieve better performance for POLYITAN-1 telemetry demodulation by scheduling only for high elevation passes at night. Moreover, due to the representation of the collected data, this study was limited to effects of the transmission environment and not on the satellite itself. This study was also unable to explore the likelihood of demodulating a signal and was restricted to the quality of the demodulation using the transfer rate. Future work could focus on the satellite itself by collecting data over a larger period and restricting observations to only high elevation night time passes study the few occurrences when the satellite is illuminated while the ground-station is not. If such a study concluded that there was a difference in transmission rate then that could be indicative of a problem with the satellite's power subsystem, such as the onboard batteries.

Appendix

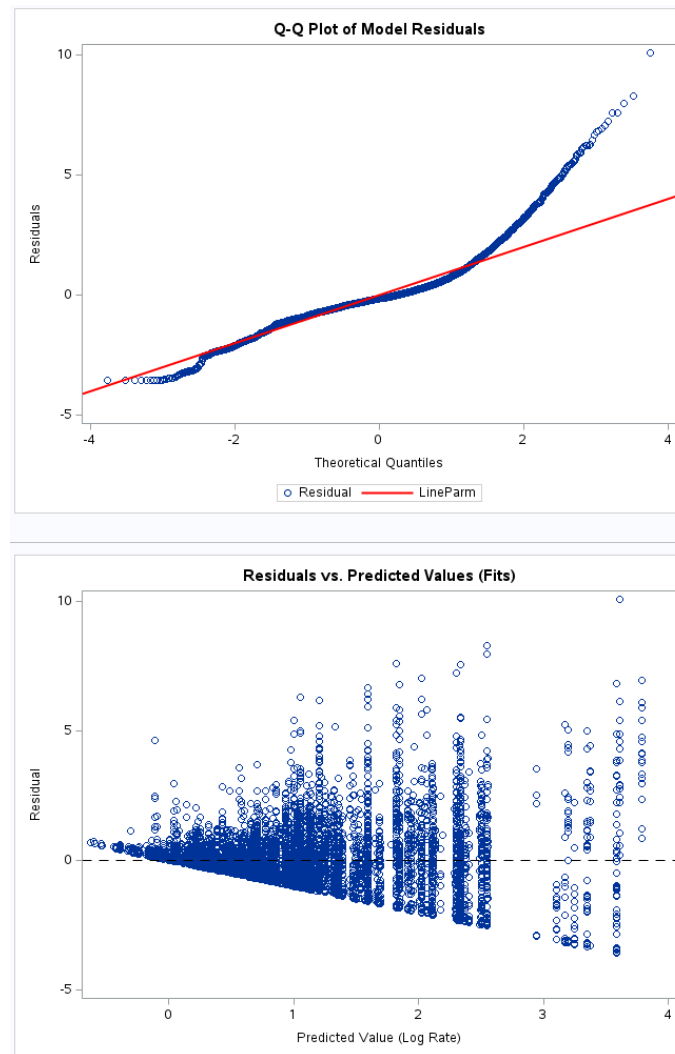


Figure 6: *Q-Q Plot and Residuals vs Fits plots showing significant violations of normalcy and constant variation of the non-transformed transmission rate.*

Type 3 Analysis of Variance								
Source	DF	Sum of Squares	Mean Square	Expected Mean Square	Error Term	Error DF	F Value	Pr > F
elevation_class	2	302.071762	151.035881	Var(Residual) + Q(elevation_class,elevation_cl*sat_lit,elevation*ground_lit,elevat*sat_li*ground)	MS(Residual)	7214	77.14	<.0001
sat_lit	1	84.288109	84.288109	Var(Residual) + Q(sat_lit,elevation_cl*sat_lit,sat_lit*ground_lit,elevat*sat_li*ground)	MS(Residual)	7214	43.05	<.0001
ground_lit	1	81.346016	81.346016	Var(Residual) + Q(ground_lit,elevation*ground_lit,sat_lit*ground_lit,elevat*sat_li*ground)	MS(Residual)	7214	41.55	<.0001
elevation_cl*sat_lit	2	13.714507	6.857253	Var(Residual) + Q(elevation_cl*sat_lit,elevat*sat_li*ground)	MS(Residual)	7214	3.50	0.0302
elevation*ground_lit	2	13.764645	6.882322	Var(Residual) + Q(elevation*ground_lit,elevat*sat_li*ground)	MS(Residual)	7214	3.51	0.0298
sat_lit*ground_lit	0	0	0	0	MS(Residual)	7214	0.00	.
elevat*sat_li*ground	0	0	0	0	MS(Residual)	7214	0.00	.
Station	75	4282.415253	57.098870	Var(Residual) + 90.179 Var(Station)	MS(Residual)	7214	29.16	<.0001
Residual	7214	14125	1.958006	Var(Residual)	.	.	.	.

Figure 7: ANOVA table showing significant co-linearity between satellite illumination and ground-station illumination.

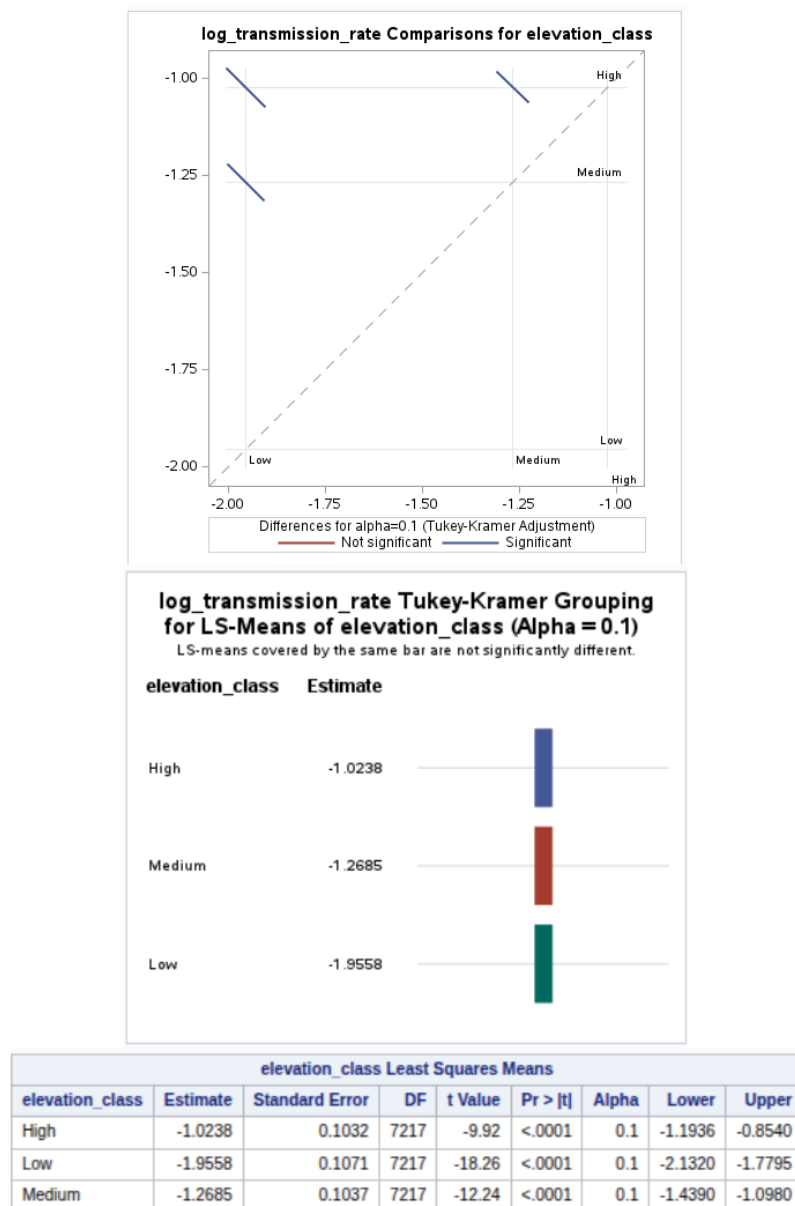
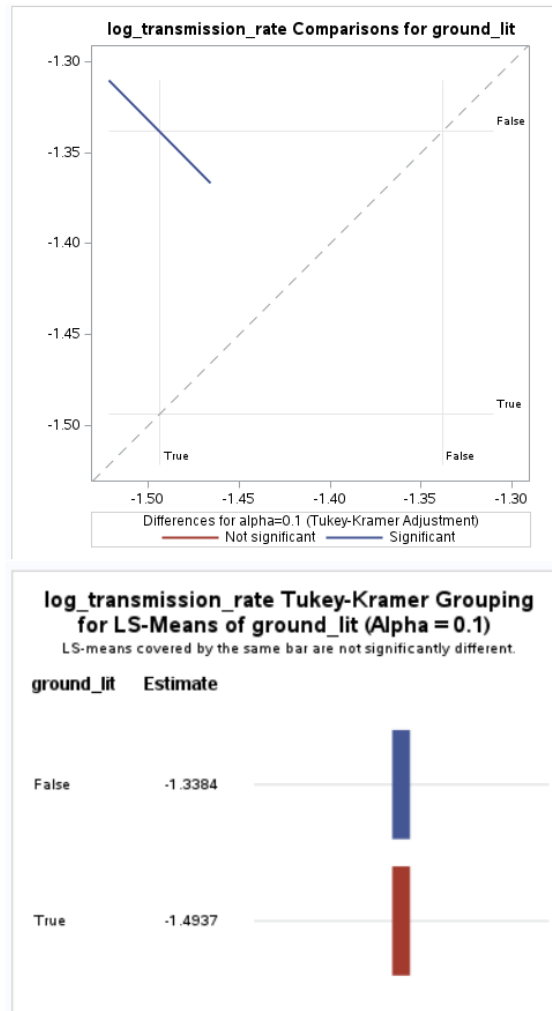


Figure 8: Diffogram and tukey grouping for elevation levels



Differences of ground_lit Least Squares Means Adjustment for Multiple Comparisons: Tukey-Kramer												
ground_lit	ground_lit	Estimate	Standard Error	DF	t Value	Pr > t	Adj P	Alpha	Lower	Upper	Adj Lower	Adj Upper
False	True	0.1553	0.03418	7217	4.54	<.0001	<.0001	0.1	0.09910	0.2116	0.09910	0.2116

Figure 9: *Diffogram and tukey grouping for ground illumination levels*

Bibliography

- [1] KPI Research Laboratory. (2015). *PolyITAN-1*. Retrieved April 13, 2026 from <http://www.cubesat.org.ua/en/nano-satellites/polyitan-1>
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- [3] Rhodes, B. (2026). *Skyfield: Elegant Astronomy for Python*. Retrieved April 13, 2026 from <https://rhodesmill.org/skyfield/>
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- [5] SatNOGS Network. (2026). *POLYITAN-1 Observations*. Retrieved April 13, 2026, from https://network.satnogs.org/observations/?sat_id=UIYC-4491-1380-9406-8215

Observations from the SATNOGS network used in this study were provided by:

- | | | |
|------------------------------|--|--|
| • 431 - PE0SAT-11 | • 2134 - MAUSyagi | • 3117 - RA3PPY |
| • 2173 - PE0SAT-21 | • 1675 - MAUSyagi | • 4372 - IK3SVW-1 |
| • 1466 - PE0SAT-12 | • 3762 - F4TNK-UHF | • 3020 - Jim UHF |
| • 1696 - PE0SAT-31 | • 6 - Apomahon | • 4151 - DEX-X |
| • 2968 - UX5UL | • 3387 - YO4DFT- S + L band | • 2416 - N5ZKK-UHF-HELICAL |
| • 3523 - IK1JNS-UHF | • 2934 - UFS-GS | • 1860 - BEEGND-4 |
| • 766 - Dunchurch | • 2510 - MAUSyagi | • 4460 - Sofia SAT Club - Bay Ivan |
| • 3538 - PU4ELT | • 2987 - VU2JEK-Nitin | • 4111 - Plyachka |
| • 2012 - SP7THR-UHF | • 754 - W6MSU UHF | • 4356 - KsienzeMale |
| • 1710 - EU1AEM | • 1155 - ArgusNavis | • 1167 - F1SXJ |
| • 2029 - EU1AEM | • 3600 - YO4DFT az 315 degrees -North West | • 3763 - plat0-3a |
| • 91 - M0EYT / 2E0NOG | • 2739 - SV5QNF - VHF | • 1002 - M0ZJO Station #1 |
| • 2729 - YL3GBC | • 3590 - EuroSpaceHub-UCM | • 4497 - VA3HEV |
| • 1864 - SA2KNG Omni UHF/VHF | • 3106 - DEX-ONE | • 4451 - I LOVE BALI YES |
| • 1936 - VK4JBE-UHF | • 3918 - YL3CT-1 | • 40 - CGBSAT-UHF |
| • 71 - PV8DX | • 2394 - YC7OIZ | • 4595 - EA4TA |
| • 3122 - Lovets | • 3951 - UT4UYF/M | • 4391 - UHF-VHF Sofia City Balcony |
| • 812 - PF_DE_UHF_X_DIPOLE | • 3905 - LW1EXU-GroundStation | • 3438 - StreetDirt UHF |
| • 1861 - LW2DYB | • 1533 - SV5QNF - UHF | • 1565 - Dumbravita - VHF/UHF - YO2MKE |
| • 2015 - Drachten | • 2247 - Mainz435 | • 1597 - YB8XM |
| • 2410 - GAO UHF | • 3925 - UT4UYF/P | • 1672 - AAS Ager Carballada |
| • 2617 - K2PI | • 4188 - ISAE.Supaero UHF | • 1523 - UHF Essendon |
| • 1376 - Venables.Shed.UHF | • 2125 - M0GLU | • 980 - PF_DE_UHF_Yagi |
| • 1416 - PE2BZ-437 | • 3508 - KEREL-UHF-RLP | • 1667 - AAS Berga Carballada |
| • 3369 - EST VENUS | • 3855 - GS ChNord UHF | |
| • 3746 - SatNOGS_Pecna.1.UHF | • 3963 - NWLGS1 | |